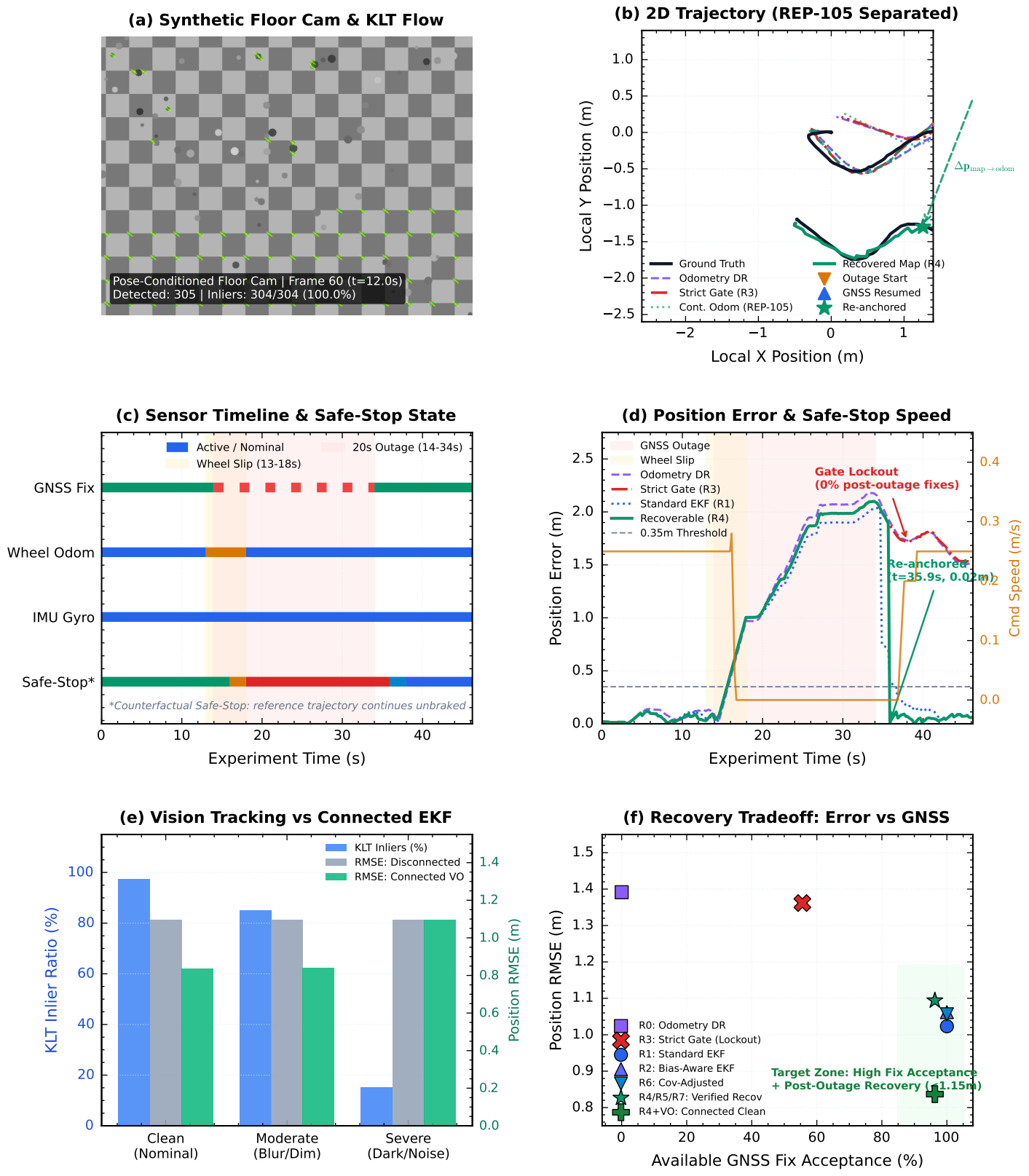

(a) Synthetic Floor Cam & KLT Flow
Pose-Conditioned Floor Cam | Frame 60 (t=12.0s)
Detected: 305 | Inliers: 304/304 (100.0%)
(b) 2D Trajectory (REP-105 Separated)
Local Y Position (m)
Local X Position (m)
Ground Truth
Odometry DR
Strict Gate (R3)
Cont. Odom (REP-105)
Recovered Map (R4)
Outage Start
GNSS Resumed
Re-anchored
$\Delta p_{map \to odom}$
(c) Sensor Timeline & Safe-Stop State
Active / Nominal
Wheel Slip (13-18s)
20s Outage (14-34s)
GNSS Fix
Wheel Odom
IMU Gyro
Safe-Stop*
*Counterfactual Safe-Stop: reference trajectory continues unbraked
Experiment Time (s)
(d) Position Error & Safe-Stop Speed
GNSS Outage
Wheel Slip
Odometry DR
Strict Gate (R3)
Standard EKF (R1)
Recoverable (R4)
0.35m Threshold
Gate Lockout
(0% post-outage fixes)
Re-anchored
(t=35.9s, 0.02m)
Position Error (m)
Cmd Speed (m/s)
Experiment Time (s)
(e) Vision Tracking vs Connected EKF
KLT Inliers (%)
RMSE: Disconnected
RMSE: Connected VO
KLT Inlier Ratio (%)
Position RMSE (m)
Clean
(Nominal)
Moderate
(Blur/Dim)
Severe
(Dark/Noise)
(f) Recovery Tradeoff: Error vs GNSS
Position RMSE (m)
Available GNSS Fix Acceptance (%)
R0: Odometry DR
R3: Strict Gate (Lockout)
R1: Standard EKF
R2: Bias-Aware EKF
R6: Cov-Adjusted
R4/R5/R7: Verified Recov
R4+VO: Connected Clean
Target Zone: High Fix Acceptance
+ Post-Outage Recovery (≤1.15m)